BIRZEIT UNIVERSITY

Electrical and Computer Engineering Department
ENCS5150 – Advanced Computer Systems Engineering Laboratory
Android Project - Digital Library Management Application
Second Summer Semester - 2025

Submission Deadline:
Discussion Date: the week after the submission deadline.
Team Work: Teams of two students
Total Marks: 100; 40% out of the final grade

Project Description:

A university library wants an Android application that allows students to browse books, reserve/borrow items, view new arrivals, and contact the library staff. The app works with online data (REST API) and also has an offline mode using SQLite. There will be two roles: Student and Librarian.

The application should include the following features:

1. **Welcome Screen Layout (5 Marks)**

- Create an animated screen that displays the library logo (use any university/library logo from the internet or create your own). Display it for 2-3 seconds with a smooth fade-in animation.
- After animation, show an Introduction layout with:
  - A brief description of the digital library system
  - A "Connect to Library" button

Connect Button behavior:

- Use REST API to fetch product categories.
  Dataset example (you can use this or create your own data or generate other random one using ChatGPT or any other method):

```
{
  "categories": [
    {"id": 1, "name": "Science & Technology", "icon_url": "..."},
    {"id": 2, "name": "Literature & Arts", "icon_url": "..."},
    {"id": 3, "name": "History & Politics", "icon_url": "..."},
    {"id": 4, "name": "Business & Economics", "icon_url": "..."}
  ]
}
```

Note: You need to replace the url with real urls.

- If the connection succeeds → navigate to Login/Registration screen.
- If it fails → show a toast message error and stay on this page.

2. Login and Registration (20 Marks)

2.1 Login (10 Marks)

- Student can enter their university ID/email and password
- "Remember Me" checkbox to save credentials in SharedPreferences
- Validate university ID format (e.g., must start with "2023" for enrollment year)
- If login succeeds → navigate to Student Dashboard
- If login fails → show appropriate error message

2.2 Registration (10 Marks)

Fields to fill:

- A Sign-Up button on login page opens a Registration layout.
- Fields:
  1. University ID (unique, must follow format: YYYY####)
  2. First Name (≥3 characters)
  3. Last Name (≥3 characters)
  4. Email (university email format: @university.edu)
  5. Password (≥6 characters, 1 uppercase, 1 lowercase, 1 number, 1 special character)
  6. Confirm Password (must match password)
  7. Academic Level (Spinner: Freshman, Sophomore, Junior, Senior, Graduate)
  8. Department (Spinner with at least 5 departments: Computer Science, Engineering, Business, Literature, Medicine)
  9. Phone Number (auto-format based on country)
- Validate inputs before submission.
- Password must be stored securely (you may use any **simple** method: eg, ceaser cipher).
- After successful registration → redirect to Login page.

3. Main Layout (10 Marks)

- Use a Navigation Drawer with:
  - Dashboard (default screen with library announcements)
  - My Borrowed Books
  - My Reading List (favorites)
  - New Arrivals
  - Profile Management
  - Library Info (Contact/Location/Hours)
  - Logout

4. Books catalog Section (20 Marks)

- Display library book collection (from API or local DB).
- Each book must show:

  1. Title
  2. Author
  3. Category
  4. Availability Status (Available, Borrowed, Reserved)
  5. Cover Image
  6. ISBN
  7. Publication Year

- Use Fragments to display books.
- Features:

  1. Search Bar: Search by title, author, or ISBN
  2. Filters: Filter by category, availability, and publication year range
  3. Add to Reading List: Save book to favorites
  4. Reserve/Borrow Button: Opens reservation form:
     - Borrowing duration (1-4 weeks)
     - Collection method (pickup from library/digital access for e-books)
     - Special notes
     - Confirm button to place reservation
  - Save order to SQLite and also sync with the server (if connected).

5. My Borrowed Books Section (5 Marks)

- Show all previous and current borrowings.
- Each entry must include:
  1. Book title and author
  2. Borrow date
  3. Due date
  4. Return date (if returned)
  5. Status (Active, Overdue, Returned, Extended)
  6. Fine amount (if overdue)

6. Reading List section (5 Marks)

- ❖ Display user's favorite/saved books
- ❖ Allow direct reservation from this list
- ❖ Show book availability status
- ❖ Remove from reading list option

7. New Arrivals Section (5 Marks)

- ❖ Display recently added books to the library collection.
- ❖ Features:
  - Add to reading list
  - Reserve/borrow directly
  - Share book recommendations

8. Profile Management (10 Marks)

- ❖ Display user information.
- ❖ View borrowing history statistics.
- ❖ User can:
  - View/update their first name, last name, password, and phone number.
  - Add profile picture.
- ❖ Password update must respect the same security conditions as registration.

9. Contact Us Layout (5 Marks)

- ❖ Three buttons:
  - **Call Library: Opens phone app with library number (+970-2-xxxxxx)**
  - **Find Us: Opens Google Maps with library location (Birzeit University Library)**
  - **Email Librarian: Opens Gmail with pre-filled email (library@birzeit.edu)**

- ❖ Additional info:

  - Library hours
  - Services offered
  - Policies and rules

10. Logout (5 Marks)

- ❖ Logs out the user and returns to the login page.

11. Librarian Dashboard(10 Marks)

- ❖ Special Navigation Drawer for librarians:

  - **Manage Students: View/delete student accounts**
  - **Add New Librarian: Register new librarian (same validation rules)**
  - **Book Management: Add, edit, delete books from catalog**

- **Reservation Management: View all reservations, approve/reject, mark as returned**
- **Generate Reports: Student activity, popular books, overdue items**
- **Library Settings: Update library information, policies**
- **Logout**

❖ To differentiate between the admin profile and the regular user, each must have a different role.

❖ The application must have a pre-created static Admin account:

Email: librarian@library.edu
Password: Library123!

Technical notes to be considered in your application:

Layouts: Dynamic RecyclerViews for book lists, static layouts for forms
Intents & Notifications: Toast messages, email intents, phone call intents
SQLite Database: Local storage for offline access, reservation cache
Animations:

- Book cover flip animation when adding to reading list
- Slide transition between fragments
- Loading spinner for API calls
- Bounce effect for reservation buttons

Fragments: Book details, reservation forms, reading lists
SharedPreferences: Login persistence, user preferences
REST API Integration: Book data, user authentication, and reservation sync
Data Validation: Form inputs, network responses, database operations
Exception Handling: Network failures, database errors, invalid inputs

Database Schema:

❖ **Books Table**: id, title, author, isbn, category, availability, cover_url, publication_year
❖ **Reservations Table**: id, student_id, book_id, reservation_date, due_date, status
❖ **Students Table**: id, university_id, first_name, last_name, email, password_hash, department, level
❖ **Reading_List Table**: id, student_id, book_id, added_date

Notes:

❖ The application must be named following the format: FirstStudentID_SecondStudentID_CourseProject.
❖ Testing will be done on a **Pixel 3a XL device with API Level 26** (Graphics = Software). Ensure compatibility.
❖ Any form of **auto-generated code using AI tools (e.g., ChatGPT) will result in a zero mark**.

- You are allowed to use internet resources for learning, but plagiarism is strictly prohibited.

What to Submit?

1. **Project.zip file** (Export from Android Studio: File → Export to zip file)
2. **APK file** (Export from Android Studio: Build → Build Bundle(s) / APK(s) → Build APK(s))

Where to submit?

Submit both the APK file and ZIP file as a reply to the message on Ritaj.

Discussion

A discussion session will be scheduled after submission. Both team members must be present and ready to explain their code, make modifications, or add additional features if requested. Not attending the discussion will result in a grade of zero.